5 (a) (i) Table 5.1 shows applications of regions of the electromagnetic spectrum.

Complete the second column of the table with the region of the electromagnetic spectrum used for each application.

Choose from the regions in this list:

gamma rays **infrared** **microwaves** **radio waves** **ultraviolet**

Each region may be used once, more than once or not at all.

Table 5.1

application	region of electromagnetic spectrum
cancer treatment	gamma rays
Bluetooth data connection	
optical fibres	
security marking	
sterilising food	
wireless internet	

[3]

(ii) State the approximate speed of radio waves in air.

speed = ... m/s [1]

(b) Fig. 5.1 shows successive crests of a wave after a plane wave has passed through a gap.

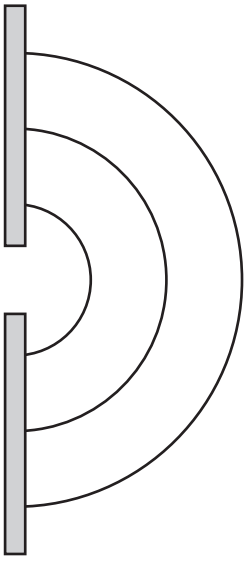


Fig. 5.1

(i) On Fig. 5.1 draw **three** successive crests before the wave reaches the gap. [2]

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(ii) Fig. 5.2 shows a much wider gap. A plane wave of the same wavelength as in **(b)(i)** is incident on the gap from the left side of the barrier.



Fig. 5.2

On Fig. 5.2, draw **three** successive crests of the wave after the wave has passed through the gap. [2]

[Total: 8]